Title

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Abstract

This is the abstract. It should be no more than 250 words and should not contain any references or displayed equations. The abstract should be a single paragraph and should not be subdivided into sections. The abstract should provide a concise summary of the paper, including the motivation, methods, results, and conclusions.

Keywords: 3 to 6 keywords, that do not appear in the title

1 Introduction

2 Literature review

2.1 Agentic data science

2.2 Communication of data analysis

2.3 AI for reproducibility

3 Method

3.1 Terminology

term	definition
decision	the unit of analysis for understanding a data analysis. It is a specific methodological choice made by the analyst that has potential theoretical alternatives. NO NEGATIVES: Do NOT create standalone decisions for negative actions (e.g., 'we did not use X').
fork (old action)	an open methodological goal that invites multiple possible approaches. It must NOT describe the final choice.
path (old decision)	a POSITIVE methodological decision that has potential theoretical alternatives, chosen to resolve the 'fork'.
justification	WHY that decision was made, extracted from the text.
tag	a unique, lowercase string starting with 'block-'.

status	should default to 'DRAFT'.

4 Application: many-analysts football study

The many-analysts study posits fixed research questions on a fixed dataset and invites teams of researchers to analyze the problem with whatever software they're comfortable with. In Silberzahn et al. (2018), one question is to answer whether soccer referees are more likely to give red cards to dark-skin-toned player than to light-skin-toned players. The provided dataset contains two raters' rating to the picture of players, the number of redcard received for each player/referee dyade out of the number of games, and a collection of player, referee, and game characteristics: player birthday, height, weight, player position, league country, number of victories, ties, and defeats, and number of yellow and red cards received. Most groups take a modelling approach to estimate the effect of skin rating on the red card issued with varied distribution assumptions (linear, logistic, and poisson) and covariate choices as fixed and random effects. The study is comprised of an initial analysis conducted separately by each team, followed by a round-robin peer evaluation round, and then each group presents their final analysis after incorporating the feedback from the discussion and presents the results in a way they would in a published article with formal Method section. All the materials related to the study is publicly available on the Open Science Framework (OSF).

From the report presented by each team, we can use LLM to convert the decisions into the schema designed and generate the corresponding coding script. From the write-up reports, we extract the sentences relevant to the decisions made in the final analysis by the team. This step is needed because the original skin rating is 1-5 but later changed to 0-1 and also

many teams report their initial analysis and how their final analysis modify on top of the initial analysis. We do this to avoid confusion on reading comprehension with LLM. For general methodology write-up, this step is not necessarily needed [TODO: write it better].

We send the text description of the extractetd decisions for LLM to generate the schema and the code script for each analysis. The code script is run by a separate LLM to validate for potential syntax errors. We manually in-place replace the `glm()` command with `fixest::feglm()` (Bergé et al. 2026) for generalized linear model and the `lme4::glmer()` (Bates et al. 2015) function with `glmmTMB::glmmTMB()` (Brooks et al. 2017) for generalized mix-effect model to speed up the evaluation of eaach script. These two functions are in-place replacements of the classic functions for estimating the relevant model with implementation with multi-threading in C++ (Bergé et al. 2026) and with Template Model Builder (TMB) (McGillycuddy et al. 2025) for better performance. This replacement should not affect the estimation of the effect size. All the LLM-related simulations are conducted with claude-opus-4.5.

In the validation phase, all team scripts were successfully executed. Notably, 10% of the scripts (n = 2) yielded successful results without requiring any automated syntax corrections by the validator, whereas the vast majority (90%) reached completion after two or three iterations. The most frequent correction involved resolving broken and overflow syntax in comments.

Figure 1 shows the agreement of the estimated effect size, in odd ratio, from our experiments (simulated) and the published effect size in Silberzahn et al. (2018) (published). For most teams, the simulation successfully recovers the odd ratio reported by each analyst team, larger variations are seen in the following teams: Team 1, 9, 15, 21, 23, and 27. Here we listed the reasons for these discrepencies upon investigation:

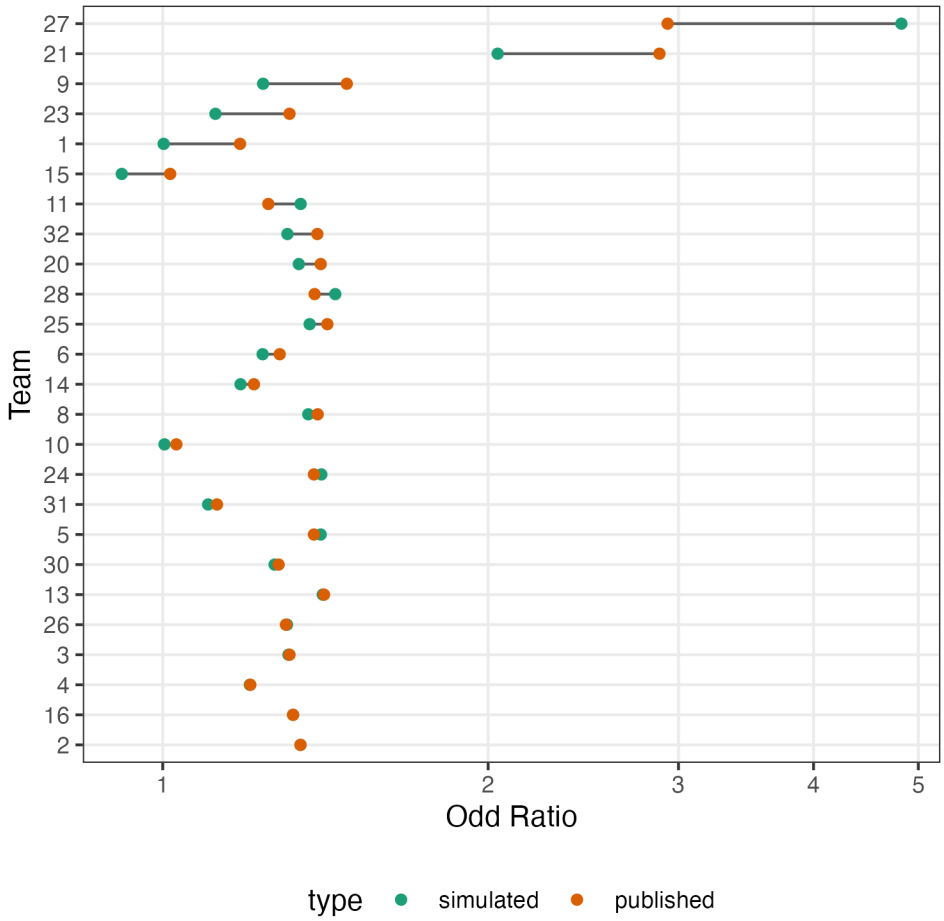


Figure 1: Comparison of the odd ratio of the effect skin color on receiving red cards in football games from our simulation with the schema (simulated) vs. the reported estimated effect size in Silberzahn et al. (2018) (published). Teams are arranged by the difference in odd ratio of our simulation and the reported value. For most teams, the simulation successfully recovers the odd ratio reported by each analyst team, larger variations are seen in the following teams: Team 1, 9, 15, 21, 23, and 27.

- **Team 27**: the diffference comes from the ambiguity of the method description, the final approach sections says *In addition to the independent variables specified in the initial approach (skin-tone rating, goals scored, games played, yellow cards, and mean explicit and implicit bias) a variable for player position was included, as well as a variable for the league of the player.* The author is intended to include the interaction effect of rating and position/ league country as other variables in the initial approach, but the LLM interprets it as only include the main effects. This error can be easily caught by comparing the reproducible script provided by the author.
- **Team 9**: the authors mistake player country for referee country in the text description, but their parameter estimation table suggests they are using player country (countryCode - Germany: France) instead.
- **Team 23 and 1**: the discrepency comes from the LLM fails to properly reshape the data from a player-referee dyad per observation to per game observation. The schema properly identifies this step but the generated code doesn't execute this step properly and create a silent error with the validator.
- **Team 15**: the discrpency comes from the default setting for estimating factor variable in R (our similuation) and SPSS (original script). In SPSS, the factor variables are estimated with effect coding, where the coefficients are compared to the grand mean, rather than to the reference group (dummy coding) in R. After changing the `contrast` options, we obtain the reported odd ratio.

4.1 Exploring alternatives

Table 2 shows …

Table 2

group	term	original	gm1	gm2	gm3
playerShort	var___(Intercept)	0.3671	0.3737	0.3738	0.2899
refNum	var___(Intercept)	0.1471	0.1528	0.1932	0.0748
refNum	var___avgrate01	NA	NA	0.0286	NA
refCountry	var___(Intercept)	NA	NA	NA	0.3990
refCountry	var___avgrate01	0.0228	NA	NA	0.0870

Appendix

4.2 Built-in prompts

Prompt: Extracting the schema from the text

```
You are an expert methodologist and data pipeline architect. I will
provide a text describing a multiverse analysis and a summary of the
actual dataset being used.


=== YOUR TASK ===


Extract a chronological list of methodological decisions (nodes) AND
map the exact data flow (inputs/outputs) for each node
simultaneously.


=== RULES ===
```

1. THEORY EXTRACTION: For each node, extract:

- 'id': A unique snake_case identifier.

- 'fork': MUST be framed as an open methodological goal that invites multiple possible approaches. It must NOT describe the final choice.

- 'path': A 'path' is strictly a POSITIVE methodological decision that has potential theoretical alternatives, chosen to resolve the 'fork'.

- 'rationale': WHY that decision (the path) was made, extracted from the text.

2. DATA MAPPING: Assign 'inputs' (EXACT column names from the dataset OR outputs from previous nodes) and 'outputs' (invented snake_case objects like 'df_clean' or 'ranef_spec').

3. ANTI-ABSTRACTION (CRITICAL): If the text lists specific variables (e.g., 'centered rater, meanIAT'), DO NOT summarize them away. You MUST capture those specific variables in the 'inputs' array.

4. INLINE ARRAYS: Format arrays strictly on one line: `inputs: [var1,

var2]`.

5. CONFIDENCE & CLARIFICATION: Rate your mapping confidence (HIGH, MEDIUM, LOW). If the text abstracts a step and you cannot confidently match it to specific dataset columns, set confidence to LOW and autogenerate a 'clarification_question' asking the user which exact columns to use. If HIGH, output 'null'.

6. Output ONLY valid YAML without markdown formatting.

=== REQUIRED YAML STRUCTURE EXAMPLE ===

meta: type: schema nodes: - fork: variables are in different scales path: apply min-max scaling to each variable justification: to put them on the same scale for combination id: step-scaling confidence: high - fork: combine the school variables into one dimension path: average exp sch and avg sch justification: the most intuitive way id: step-education confidence: low

=== DATASET SUMMARY ===

NA

=== METHODOLOGY TEXT ===

```
[inset text here]
```

Prompt: generating code from the schema

```
You are an expert R programmer. Attached is a text document
containing a SCHEMA that defines a data processing pipeline.

No data file is specified - if data loading is required, use a
sensible placeholder or note it in a comment. Your task is to write a
complete, working R script that implements this pipeline
step-by-step. Use modern R practices (like dplyr or base pipe) and
ensure variables flow correctly from one step to the next as defined
by the inputs and outputs. Output ONLY the complete R script. Do not
start with markdown formatting blocks (like ```R) or backticks.
```

Prompt: validating the code generated

```
You are an expert R developer and automated debugging agent. The user
has provided a data file located at the relative path: `data/[insert
the path to the data]'. Your purpose is to receive broken R scripts,
diagnose the execution error, and return the corrected script.

CRITICAL INSTRUCTIONS: 1. Fix the bug while preserving the original
intent and logic of the script.  2. If the error is a missing
package, add the necessary `library()` call at the top.  3. If the
error is a missing variable, ensure it is properly initialized before
```

```
use.  4. Do NOT hallucinate new data files or external dependencies
unless explicitly provided in the context.

STRICT OUTPUT CONSTRAINTS: - You must output ONLY valid, fully
executable R code.  - Absolutely NO markdown formatting. Do NOT wrap
your response in ```R or ```.  - Absolutely NO conversational filler,
greetings, explanations, or comments about what you fixed.  - The
first character of your response must be R code, and the last
character must be R code.
```

Prompt: generating alternatives for a specific step

```
You are an expert Data Analyst and Methodologist. You are reviewing
an analysis schema to identify "Forking Paths" -- alternative
analytical choices that are equally valid but distinct from the
current approach.

=== DEFINITIONS ===

The schema provided to you consists of steps with:

- **ACTION**: The goal of the step (What needs to be done).

- **DECISION**: The specific implementation chosen (How it is done).

- **JUSTIFICATION**: The reasoning behind that decision.
```

- **ID**: The unique identifier for the step (kebab-case).

=== TASK ===

Focus specifically on the step tagged: "[inset step number here]". Your goal is to generate 3 distinct, valid alternatives for this step.

For each alternative: 1. **Keep the same ACTION** (the goal remains constant).

2. **Change the DECISION** to a different but methodologically sound approach.

3. **Provide a new JUSTIFICATION** explaining why this alternative is valid.

4. **Create a new ID** that reflects the new decision (must be kebab-case).

=== OUTPUT FORMAT ===

Please output the result in **strictly valid YML format**.

Crucial Formatting Rules:

1. Include a `meta` section at the top with `type: alternative` and the `step`.

2. Output strictly valid YML. All text values (decision, justification) must be enclosed in double quotes ("). Do not use block styles (| or >). Do not wrap lines or insert \n characters within the quotes; keep the text as a single continuous string.

3. Do not include markdown code fences (like ```yml) or conversational text. Just the raw YML.

=== REQUIRED YML STRUCTURE EXAMPLE ===

meta: type: tines_alternative step: [inset step number here] alternatives: - id: step-new-method-name action: Repeat the original action decision: "Description of the new decision..." justification: "This is the reasoning for why this alternative is valid." - id: step-another-method ...

References

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